

PawWatch

Edge-AI Cat Health Monitor - 2026.10.10

Jin-Yu Yang, Zan-Yun Lee and Wen-Hsiu Lin

Department of Computer Science and Information Engineering, National Taiwan University

Problem

Cat hides illness

Cats instinctively hide pain and illness; obvious symptoms such as vomiting may appear only when it is more severe [1], [2]. Earlier signs show up in daily habits: vets list changes in appetite, thirst and litter box use as warning signs [2], and each common disease changes them in its own way:

Disease	How common	Eating	Drinking	Litter
Kidney disease	Up to 30–40% of older cats may have it; only 3.6% are diagnosed [3], [4]		↑	↑
Hyperthyroidism	8.7% of cats over 10 [5], [6]	↑	↑	↑
Diabetes	About 1 in 200 cats [7], [8]	↑	↑	↑
Urinary tract disease	A leading reason cats visit vets [9]			↑

A 2026 study of 136 cats found those with kidney disease used the litter box about twice as often, even in early stages [10]. A blocked urinary tract may kill within 24–48 hours; its first sign is many litter box trips [9].

Solution

Today's smart devices each watch one spot

Watching food, water and litter today takes several devices, brands and apps.

	Food	Water	Litter	Video home
Pet cameras (Furbo, Petcube) [11], [12], [13]	✗	✗	✗	✗
Smart litter boxes (Petivity, Litter-Robot, PETKIT) [14], [15], [16]	✗	✗	✓	—
Smart fountains (Felaqua Connect) [17]	✗	✓	✗	—
Smart feeders (Petlibro Granary 2) [18]	✓	✗	✗	✗
PawWatch	✓	✓	✓	✓

“Video home” = video stays at home

However, illness shows up as a **combination**. For example, kidney disease raises both drinking and litter box use [3], so a water-only or litter-only device sees half the pattern.

Solution: PawWatch

Owners are out all day, and no single device sees the whole pattern. **PawWatch** is a vision-based monitoring system: low-cost cameras watch the food bowl, water bowl and litter box, and an on-device AI accelerator recognizes the cat and logs every visit. Video never leaves home.



What it records

- **Visit log:** how often and how long the cat eats, drinks and uses the litter box
- **No-go zone alarm:** plays a sound when the cat jumps onto the counter

What it tells you

- **Anomaly alert:** ≥ 2 or $\leq \frac{1}{2}$ times of the cat's 7-day average
- **Daily report:** a plain-language summary
- **Dashboard:** today's counts, timeline and 7-day trends

Why run AI at home, not in the cloud

	Cloud pet camera	PawWatch
Who sees your home video?	It's uploaded to the company's servers. At one home-camera maker, staff watched customers' private videos. [19], [20]	No one. Video never leaves home; only events like "14:32, drank for 25 s" are stored
Monthly fee	Smart features need a plan, typically US\$60–120/yr [12], [18]	None. No servers to pay for
Internet goes down	Some apps stop updating [16]	Keeps logging; alerts are sent once it's back
Cat jumps on the counter?	Frames go to the server and back first	The alarm sounds right away, from the box at home

Running 24/7 is cheap: the AI accelerator draws 2.5 W, less than an LED bulb [21].

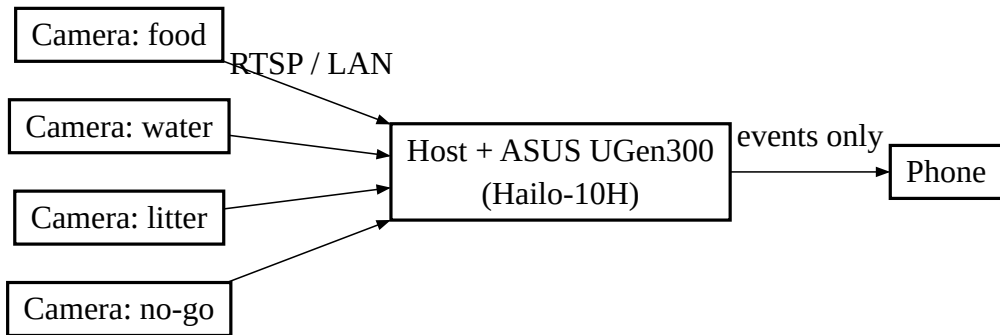
PawWatch System

Architecture

TODO: system architecture diagram and technology overview

Hardware: one host, many fixed cameras

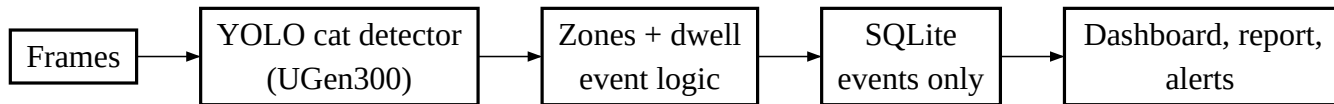
TODO: 目前是 AI 生的



- Fixed cameras: zones are drawn once; the cat is large in frame
- UGen300 is an M.2 / USB card in the host, not a standalone computer

Software: video in, events out

TODO: 目前是 AI 生的



- Detector contract (label, score, box): laptop backend today, Hailo backend is a swap
- Stock COCO YOLOv8, precompiled for Hailo-10H: no training, no conversion
- Frames are discarded after inference

The load fits the chip

TODO: 目前是 AI 生的

Needed	3 cameras $\times$ 5 fps = 15 inferences/s
Laptop (M3 Pro CPU, YOLOv8n 640)	49 fps measured
Hailo-10H (YOLOv8n)	TODO: published fps, cite source

TODO: Stage II plan: port to UGen300, measure fps and power

Impact and Business

Expected benefits

TODO: 目前是 AI 生的

- **Owners:** catch changes early instead of when symptoms are obvious
- **Vets:** objective at-home records after surgery or before a follow-up

Without PawWatch

The cat uses the litter box twice as often while the owner is at work. Nobody notices until symptoms are obvious.

With PawWatch

Litter box visits reach $2\times$ the 7-day average: an alert the same day suggests watching the cat or seeing a vet, with the daily report in hand.

Outcomes:

- Stage I, a working prototype on recorded video and open-source code on GitHub;
- Stage II, the detector on the **UGen300** with measured fps and power.

Market: more cats, older cats, growing pet tech

TODO: 目前是 AI 生的, 感覺留下 more cats?

- **Taiwan:** 1.74 M cats, +33% in two years, now more than dogs [22]
- **Older cats:** Japan's average cat lifespan hit 15.79 years [23]
- **Pet tech:** US\$15.6 B (2025) → US\$52.9 B (2035), 12% a year; health is the largest segment [24]

	Who	Value / yr
TAM	Global pet tech, health segment [24]	US\$4.4 B
SAM	Taiwan cat homes with a cat 7+: ≈ 460 k	NT\$4.6 B
SOM	1% of SAM in 3 years: $\approx 4,600$	NT\$46 M

Assumptions: 1.5 cats per home → 1.16 M homes [22]; 40% have a cat 7+ (Japan proxy, 44%) [25]; NT\$10k per home (assumed); US\$1 $\approx$ NT\$32.

Next: Taiwan → Japan (9 M cats) [23] → US (53 M cat households) [26]

Business model: ride on hardware people already buy

TODO: 目前是 AI 生的, 完全不知道這啥

Customer	Offer	Revenue
Device makers (pet cameras, home hubs, PCs with edge AI)	PawWatch edge software on their hardware: no cloud cost for them	Per-unit license
Vet clinics	At-home monitoring kits after surgery and for chronic cases (e.g. kidney disease)	Monthly rental per kit
Cat owners	Box + cameras, no subscription	One-time purchase

- Licensing spreads the UGen300 cost: it doesn't land on one product
- Today's kit: UGen300 US\$299 + host + $3 \times \sim\text{US\$}20$ cameras $\approx \text{US\$}430\text{--}500$ [21], [27], [28], [29]

What's next: from prototype to product

TODO: 目前是 AI 生的

Stage I (now)

- Laptop prototype on recorded video
- Visit log, anomaly alert, daily report, dashboard, no-go alarm

Stage II (11/6–12/4)

- Port the detector to the UGen300
- Measure fps and power
- On-device LLM daily report

After the contest

- Pilot with real cats at home
- Multi-cat identification
- Pan-tilt cameras; auto-detect bowls and litter box
- Street-cat TNR tracking (ear tips)

Code and references

TODO: 其實應該可以全部丟 Appendix 吧

Code: github.com/illumew/PawWatch

References: numbered markers such as [1] on each slide refer to the full list in the appendix: veterinary research and guidelines, product and price pages, and market data.

Appendix

Other

感覺能放一些截圖？或是前面放不下的圖表 blah blah blah

References

- [1] M. C. Evangelista et al., “Facial expressions of pain in cats: the development and validation of a Feline Grimace Scale,” *Scientific Reports*, 2019, [Online]. Available: <https://ucalgary.ca/news/cats-faces-reveal-their-hidden-pain>
- [2] VCA Animal Hospitals, “Recognizing Signs of Illness in Your Cat.” [Online]. Available: <https://vcahospitals.com/resources/conditions-cat/recognizing-signs-of-illness-in-your-cat>
- [3] A. H. Sparkes et al., “ISFM Consensus Guidelines on the Diagnosis and Management of Feline Chronic Kidney Disease,” *Journal of Feline Medicine and Surgery*, 2016, [Online]. Available: <https://journals.sagepub.com/doi/10.1177/1098612x16631234>
- [4] M. Conroy, D. C. Brodbelt, D. O'Neill, Y. Chang, and J. Elliott, “Chronic kidney disease in cats attending primary-care practice in the UK: a VetCompass study,” *Veterinary Record*, 2019, [Online]. Available: <https://rvc-repository.worktribe.com/OutputFile/1382711>
- [5] M. J. Stephens et al., “Feline hyperthyroidism reported in primary-care veterinary practices in England,” *Veterinary Record*, 2014, [Online]. Available: <https://rvc-repository.worktribe.com/output/1404762>
- [6] M. E. Peterson, C. A. Castellano, and M. Rishniw, “Evaluation of body weight, body condition, and muscle condition in cats with hyperthyroidism,” *Journal of Veterinary Internal Medicine*, vol. 30, no. 6, pp. 1780–1789, 2016, [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5115195/>
- [7] D. G. O'Neill et al., “Epidemiology of diabetes mellitus among 193,435 cats attending primary-care veterinary practices in England,” *Journal of Veterinary Internal Medicine*, 2016, [Online]. Available: <https://www.rvc.ac.uk/vetcompass/news/epidemiology-of-feline-diabetes-mellitus>
- [8] Merck Animal Health, “Diabetes in cats: FAQ.” [Online]. Available: <https://www.merck-animal-health-usa.com/mkt/pet-diabetes-month/cats-faq-2>
- [9] Cornell Feline Health Center, “Feline Lower Urinary Tract Disease.” [Online]. Available: <https://vet.cornell.edu/departments-centers-and-institutes/cornell-feline-health-center/health-information/feline-health-topics/feline-lower-urinary-tract-disease>

References (2)

- [10] Langenfield-McCoy et al., “Enhancing Detection of Chronic Kidney Disease Through Smart Litter Box Monitoring,” *Animals*, vol. 16, no. 9, p. 1319, 2026, [Online]. Available: <https://www.mdpi.com/2076-2615/16/9/1319>
- [11] Furbo, “Furbo 360° Cat Camera.” Accessed: Oct. 10, 2026. [Online]. Available: <https://furbo.com/us/products/furbo-360-cat-camera>
- [12] ThePricer, “How much does a Furbo subscription cost?.” [Online]. Available: <https://www.thepricer.org/how-much-does-furbo-subscription-cost/>
- [13] Petcube, “What is Petcube Care membership.” [Online]. Available: <https://petcube.com/support/article/petcube-care-membership/>
- [14] Petivity (Purina), “Petivity product list.” [Online]. Available: <https://www.petivity.com/llms.txt>
- [15] Whisker, “Litter-Robot 4 Insights Bundle.” [Online]. Available: <https://www.whisker.com/litter-robot-4-insights-bundle>
- [16] PETKIT, “Purobot Ultra product introduction.” [Online]. Available: https://ipet.ch/media/file_attachments/attachment/PETKIT_PUROBOT_ULTRA_-_Product_Introduction.pdf
- [17] Sure Petcare, “Felaqua Connect.” [Online]. Available: <https://www.surepetcare.com/en-us/felaqua-connect>
- [18] TechCrunch, “Petlibro's new AI-powered feeder is a game changer for multi-cat homes.” [Online]. Available: <https://techcrunch.com/2026/10/06/petlibros-new-ai-powered-feeder-is-a-game-changer-for-multi-cat-homes/>
- [19] M. Ehrenkranz, “How To (Responsibly) Spy On Your Pets,” Mozilla Foundation. [Online]. Available: <https://www.mozilla.org/en/blog/how-responsibly-spy-your-pets>
- [20] Malwarebytes, “Ring agrees to pay 5.6 million after cameras were used to spy on customers”. [Online]. Available: <https://www.malwarebytes.com/nl/blog/news/2024/04/ring-agrees-to-pay-5-6-million-after-cameras-were-used-to-spy-on-customers>
- [21] iThome, “ASUS UGen300 press release (NT6,380, 2.5 W, 40 TOPS)”. [Online]. Available: <https://www.ithome.com.tw/pr/175676>
- [22] PTS News, “Ministry of Agriculture 2025 household dog and cat survey.” [Online]. Available: <https://news.pts.org.tw/article/804498>
- [23] Asia News Network, “Japan's ageing cats feeling paw-sitive effects of AI healthcare.” [Online]. Available: <https://asianews.net>

References (3)

- network/japans-ageing-cats-feeling-paw-sitive-effects-of-ai-healthcare
- [24] Global Market Insights, “Pet Tech Market report.” [Online]. Available: <https://gminsights.com/industry-analysis/pet-tech-market>
- [25] GlobalPETS, “Japanese pet market.” [Online]. Available: <https://globalpetindustry.com/article/japanese-pet-market/>
- [26] PetfoodIndustry, “Report: Cat ownership grows to 53 million U.S. households (APPA 2026 Cat Report).” [Online]. Available: <https://www.petfoodindustry.com/pet-food-market/market-trends-and-reports/news/15834866/report-cat-ownership-grows-to-53-million-us-households>
- [27] SHI, “ASUS UGen300 USB AI module (Hailo-10H, 8 GB).” [Online]. Available: <https://texas.gs.shi.com/product/52158156/ASUS-UGEN300-USB-AI-MODULE-HAILO-10H-8-GB-LPDDR4-USB-3.1-GEN2>
- [28] Notebookcheck, “Raspberry Pi 5 now costs up to 205 due to RAM crisis”. [Online]. Available: <https://www.notebookcheck.net/Raspberry-Pi-5-now-costs-up-to-205-due-to-RAM-crisis.1218213.0.html>
- [29] Slickdeals, “TP-Link Tapo C110 indoor Wi-Fi camera, 4-pack deal.” [Online]. Available: <https://slickdeals.net/f/17777847-4-pack-tp-link-tapo-c110-3mp-indoor-wi-fi-security-camera-with-night-vision-50-free-s-h-49-99>